# FVS-ACD and OSM, head to head across the Acadian region

## Benchmark status, direct comparison, and the path to v13.1

Canadian Forest Service, Acadian Growth & Yield Meeting | 18 August 2026

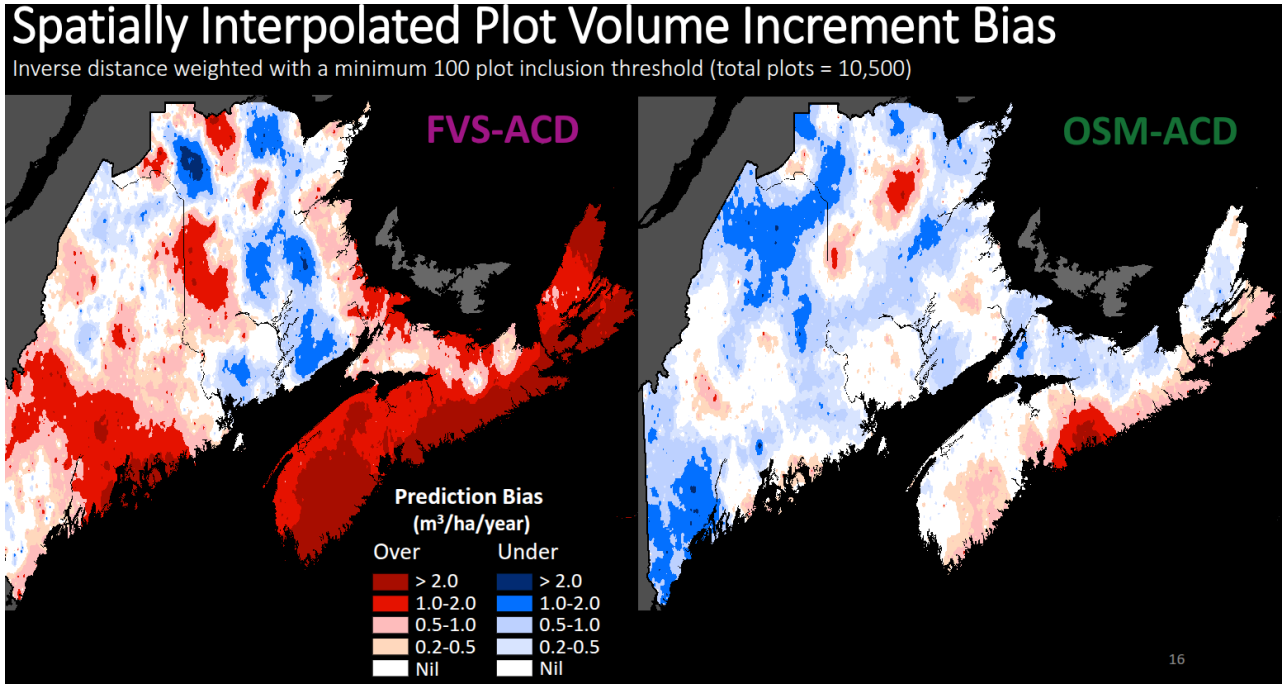
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*FVS-ACD v12.11 wins on managed and mature stands; OSM keeps the lower pooled error overall. v13.1 targets what is left.*

CONTEXT

# Two Acadian growth models, one shared benchmark

**2018 ECANUSA: OSM edged FVS-ACD on merchantable volume increment**
**2026: v12.10.1 ships, one bias survives**

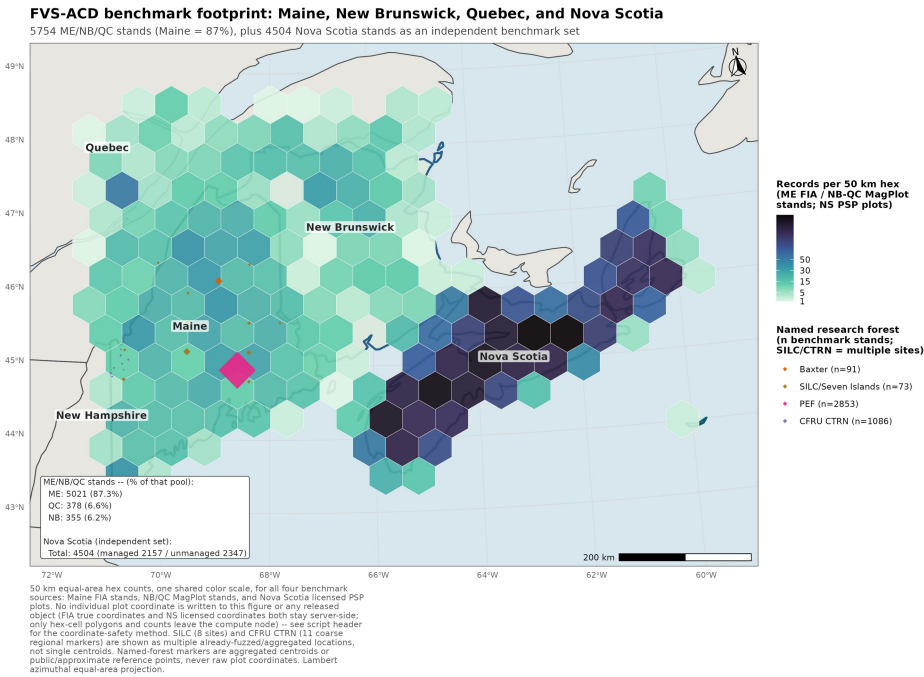


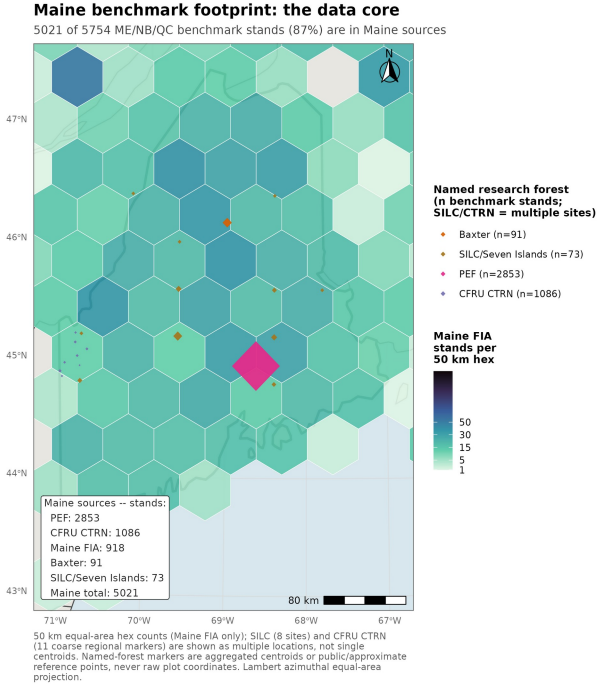

1. One benchmark: eight sources, one volume currency
2. Head to head: FVS-ACD v12.11 versus OSM v2.26.1
3. The maturity bias, and what v12.11 fixed
4. Independent test: Nova Scotia
5. Next: v13.1 (ship18), scored and disclosed

*What this talk covers*

DATA FOOTPRINT

# Maine supplies 87.3% of the benchmark, Nova Scotia is independent




**5,021 / 5,754**
Maine stands, 87.3% of the pooled ME/NB/QC total

**355**
New Brunswick stands

**378**
Quebec stands

**4,504**
Nova Scotia stands, managed and unmanaged (independent test)

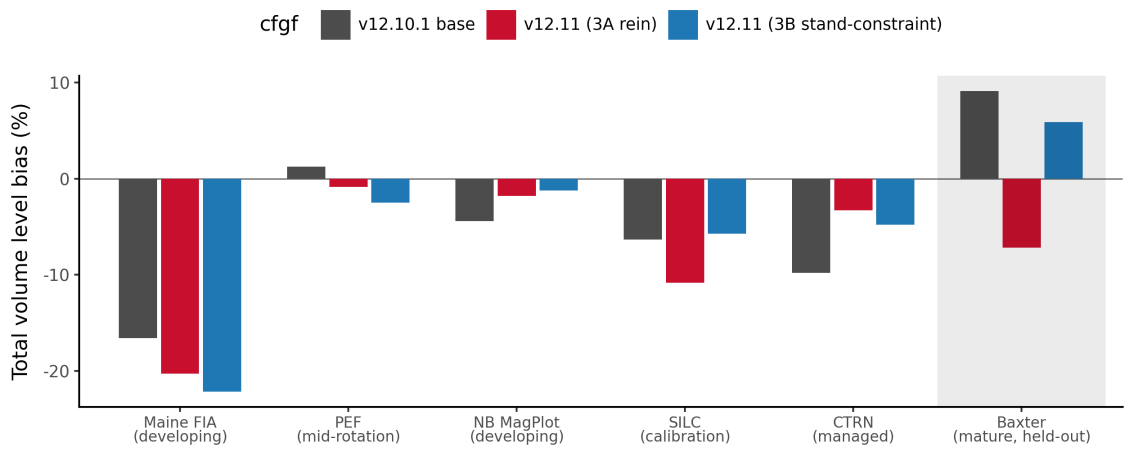
**8**
sources across four jurisdictions

*Nova Scotia never used to fit or tune any lever. Seven pooled ME/NB/QC sources on one NSVB currency: PEF 2,853, CTRN 1,086, Maine FIA 918, Baxter 91, SILC 73, NB 355, QC 378; Nova Scotia is the eighth, scored separately.*

THE PROBLEM

# Volume bias flips sign across the maturity gradient

Total volume bias




Basal area bias

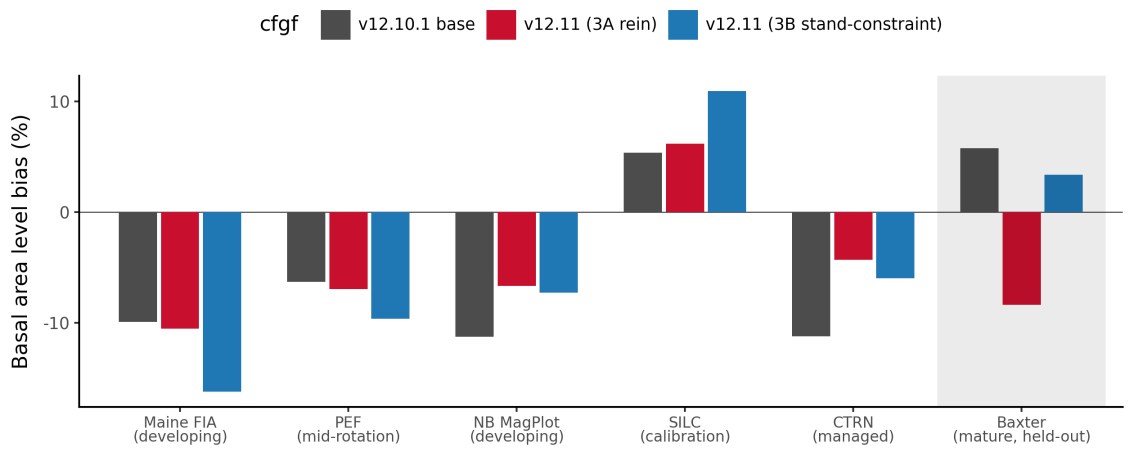



*Figure 1. Total volume and basal area level bias across the stand-maturity gradient, base v12.10.1 versus the v12.11 candidate. Baxter panel predates the 15 Aug survivor-matching fix; corrected Baxter levels are base +13.6, candidate +11.3, OSM +16.9 (ordering unchanged).*

- Base to v12.11 candidate, versus OSM v2.26.1
- Maine FIA developing -16.60% to -16.13% (OSM -6.29%)
- CTRN managed -9.78% to -2.24% (OSM -4.81%)
- Baxter mature, held out +13.56% to +11.28% (OSM +16.85%)

CENTRAL RESULT

# On its target managed and mature sources, v12.11 is similar or better than OSM v2.26.1

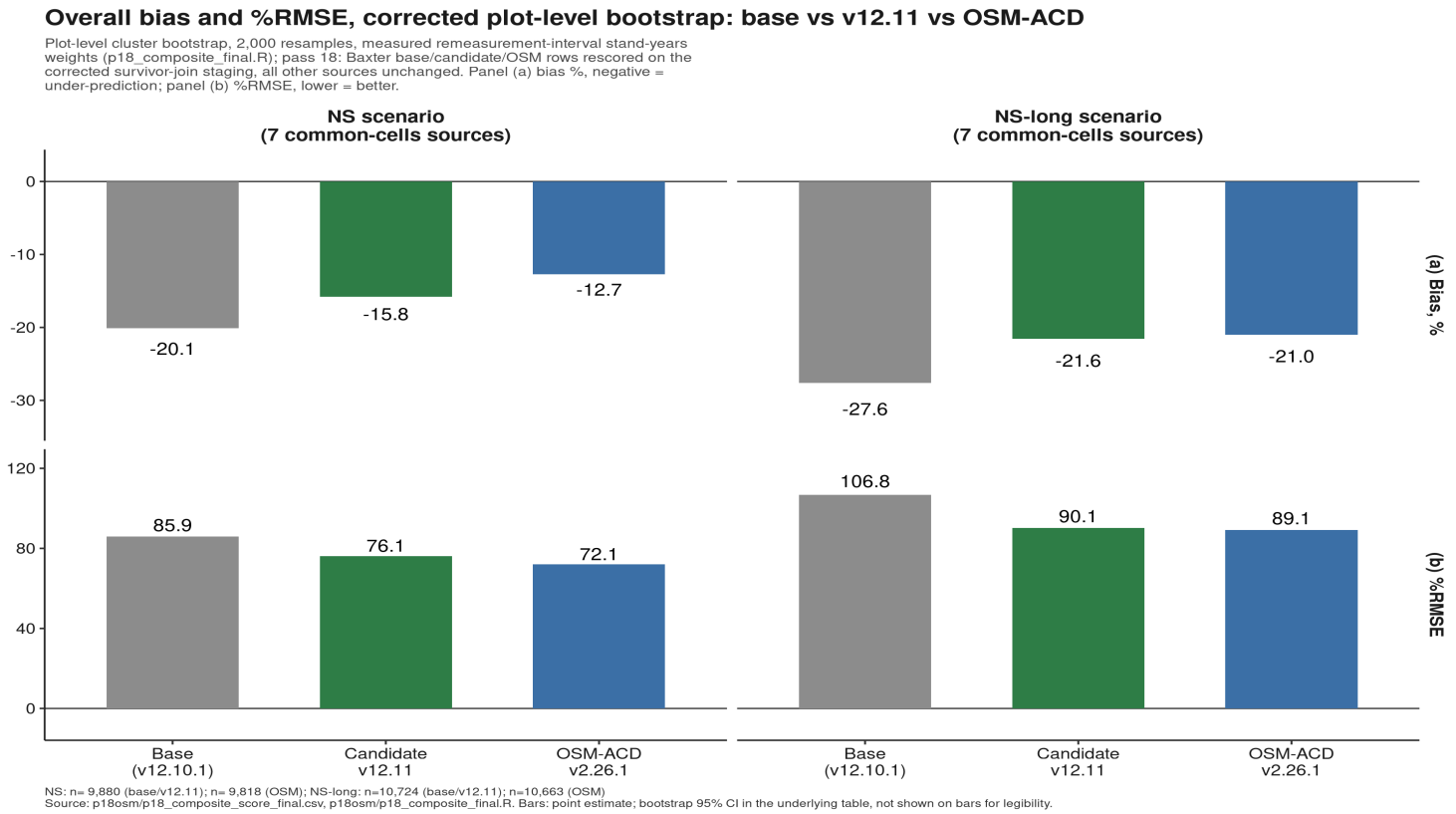
| Source | n | Total volume bias base / v12.11 / OSM | Basal area bias base / v12.11 / OSM |
|---|---|---|---|
| Baxter, ME (mature, held out) | 91 | +13.56% / *+11.28%* / +16.85% | +7.63% / *+6.34%* / +15.36% |
| CTRN, ME (managed) | 1,086 / 1,033 | -9.78% / *-2.24%* / -4.81% | -11.26% / *-3.52%* / -4.19% |
| NB MagPlot (managed) | 355 | -11.84% / *-5.96%* / -8.07% | -8.84% / *-2.33%* / -2.48% |
| Overall | 1,532 / 1,479 | -2.68% / 1.02% / 1.32% | -4.15% / 0.16% / 2.89% |

*All three sources scored at stand grain. Baxter reflects the 15 Aug survivor-matching data fix (all three models rescored on corrected observed data; ordering unchanged, RMSE improves to 28.7 / 28.5 / 32.2). QC MagPlot has no scored OSM comparison yet -- the input was never built, not a vendor-access limit; a working, licensed OSM-ACD v2.26.0 binary already runs on this project's own Cardinal account, with prior runs on several of this project's sources. OSM still leads developing Maine FIA at either grain (subplot gap moves only 0.08 points) -- the acknowledged v13.1 target.*

OVERALL SUMMARY

# Pooled across everything: OSM is ahead, v12.11 is close, base is behind

**NS scenario (7 sources): %RMSE 85.9 base / 76.1 v12.11 / 72.1 OSM (bias -20.1 / -15.8 / -12.7). NS-long scenario (7 sources): %RMSE 106.8 base / 90.1 v12.11 / 89.1 OSM (bias -27.6 / -21.6 / -21.0).**

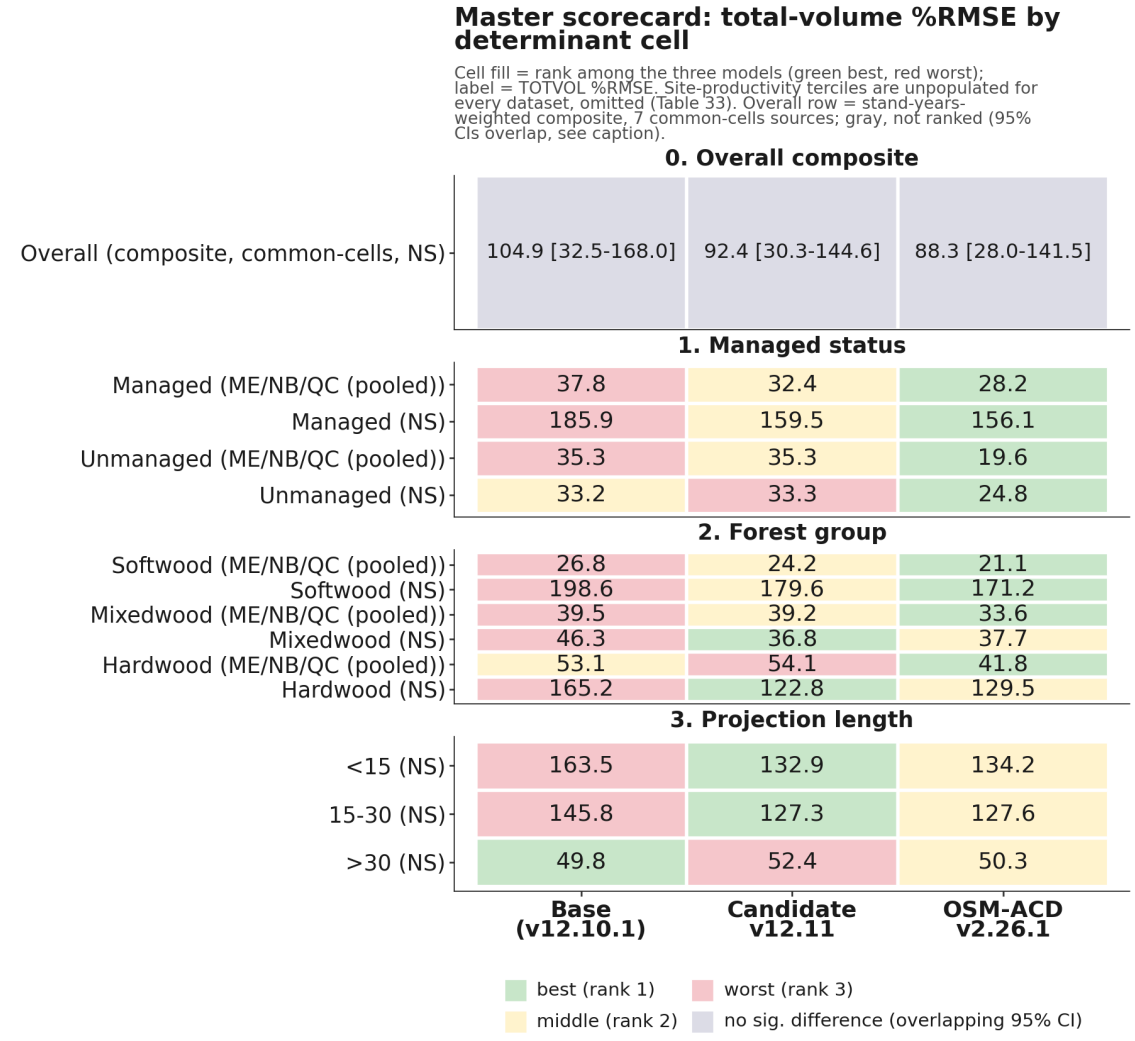



*Error pooled across all seven shared benchmark sources, weighted by stand-years, with bootstrap uncertainty intervals. v12.11 and OSM sit inside each other's intervals under both scenarios; the frozen base is clearly behind on the long-horizon scenario. Values refreshed 15 Aug after a benchmark data fix.*

MASTER SCORECARD

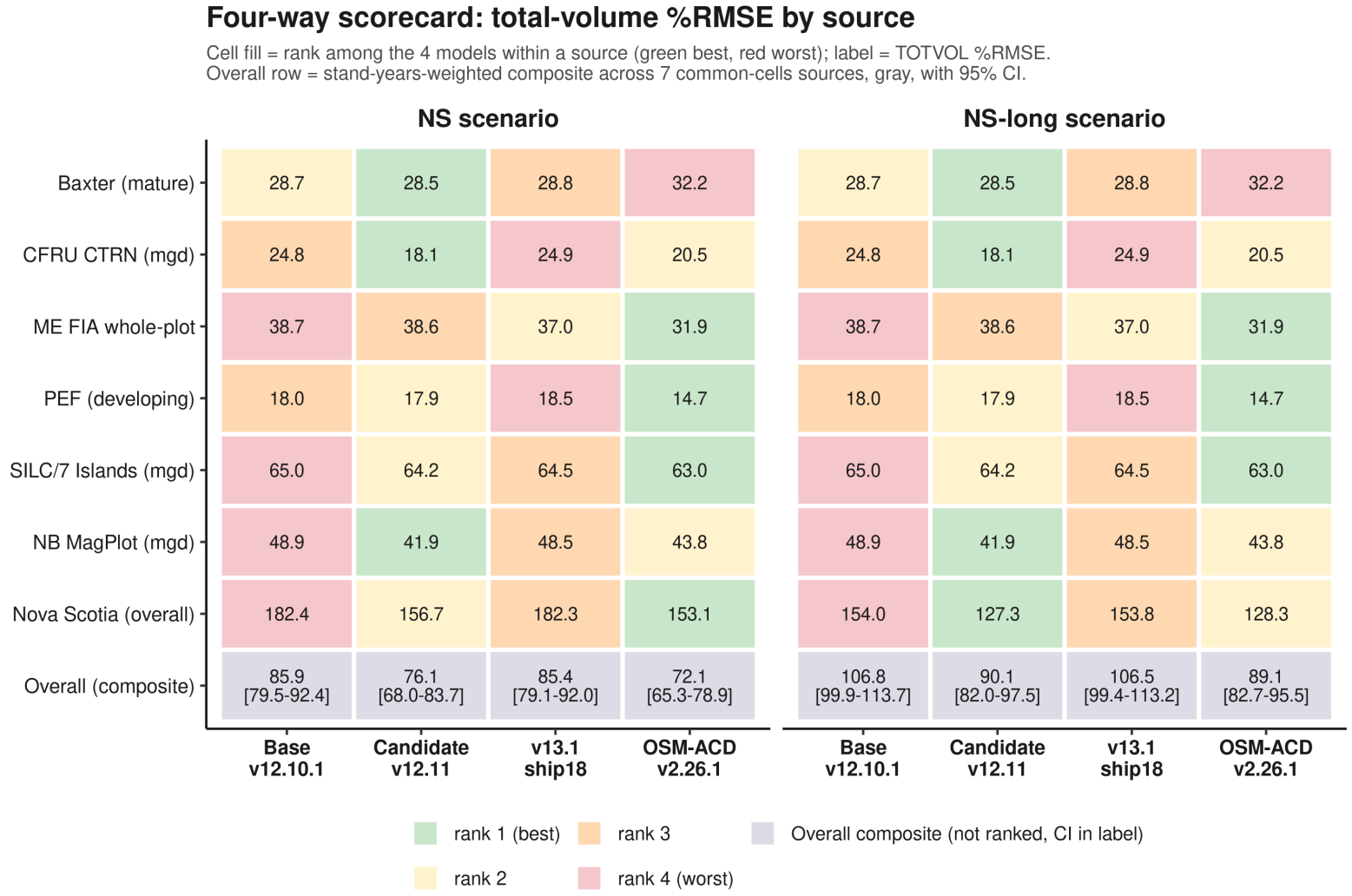
# OSM leads managed; v12.11 wins Nova Scotia hardwood and mixedwood

*Overall composite row above predates the plot-level bootstrap correction and the 15 Aug Baxter data fix; corrected values (Overall Summary, slide 9): NS 85.9/76.1/72.1, NS-long 106.8/90.1/89.1 (base/v12.11/OSM). OSM's pooled ME/NB/QC cells include the three sources touched by the fix; Maine FIA and SILC verified unchanged, Baxter rescored, no ordering flips.*



MASTER SCORECARD, FOUR-WAY

# v13.1 (ship18): fixes its target gap; the overall composite still trails

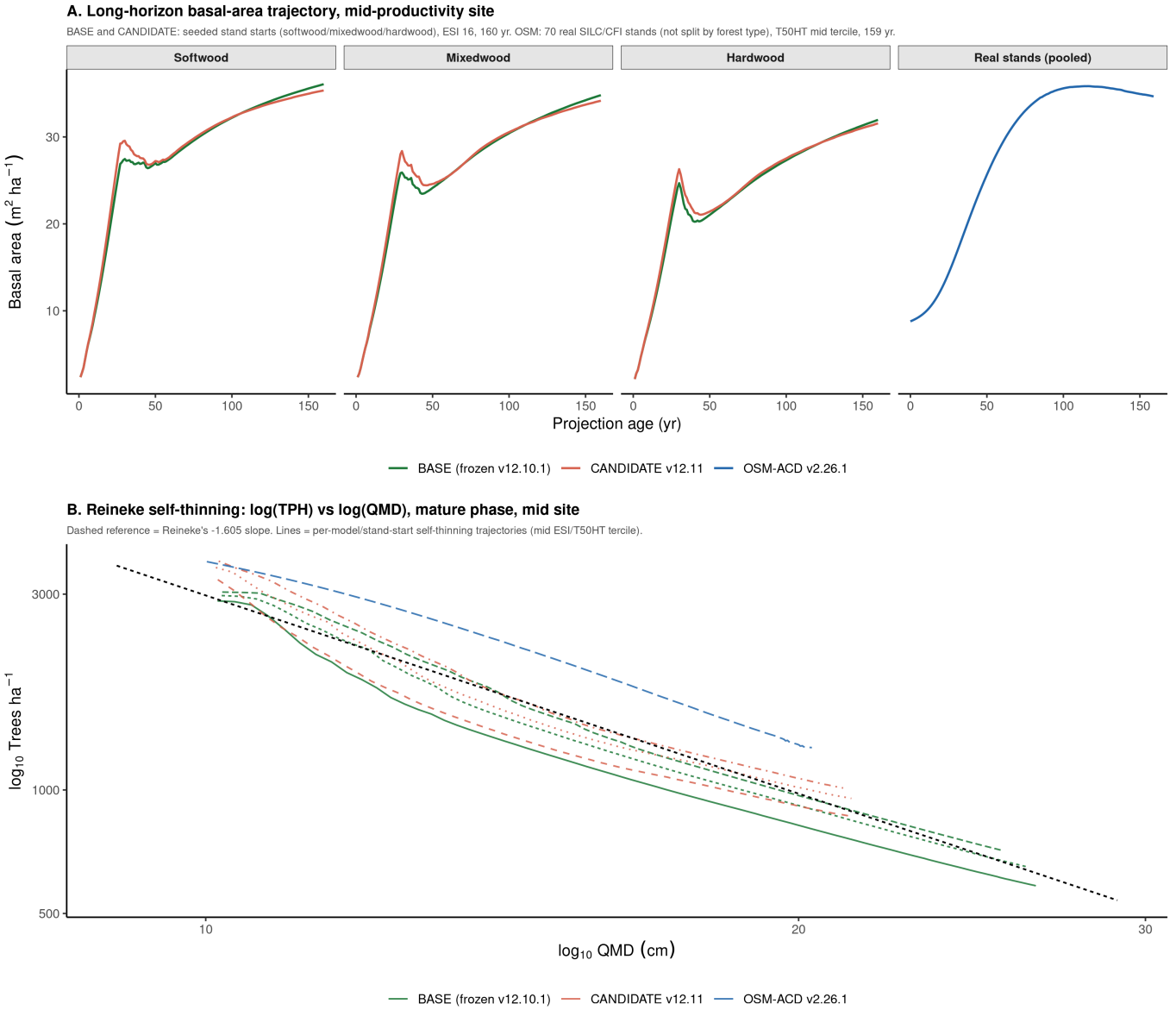



**ship18 is v13.1's shipping candidate: a stand-level survivor-growth constraint, higher ingrowth, a refit diameter increment, and a retuned density limit, each off by default. It fixes what it targets: developing Maine FIA, the gap v12.11 left open, improves from -16.6% to -12.5% bias (error 38.7 to 37.0), with other sources essentially unchanged. Pooled across all seven sources it scores 85.4 (NS) and 106.5 (NS-long), essentially tied with the frozen base and behind v12.11 (76.1 / 90.1) and OSM (72.1 / 89.1), because Nova Scotia dominates the pooled weight and ship18 does not change Nova Scotia. Disclosed tradeoffs: one self-thinning check lands exactly on the acceptance boundary (Reineke softwood -1.80), and small penalties at PEF and Baxter.**

LONG-HORIZON BIOLOGICAL REALISM

# Do the models stay biologically sensible over 160 years? FVS-ACD: 4 of 5 checks; OSM: 3 of 5

- *Eichhorn: v12.11 pass, OSM flag, CV 26.3%*
- *Site ordering: v12.11 a brief crossing, OSM a persistent inversion*
- *OSM passes 3 of 5 (fails Eichhorn, CV 26.3%, and site ordering).*



NOVA SCOTIA

# Independent test: Nova Scotia. v12.11's gains hold; neither model is uniformly better

| | | |
|---|---|---|
| 1 | **4,504 scored pairs, fully independent** | Nova Scotia PSP network, established 1965<br>Never used to fit v12.11 |
| 2 | **Managed total volume bias -58.10% to -42.32%** | A 27% relative reduction |
| 3 | **OSM head to head, and the integrity check** | Managed: essentially tied (v12.11 -42.32% vs OSM -41.96%). Long horizon (>30 yr) is the clearest separation: v12.11 -1.66% vs OSM -9.14%. Site index confirmed on the same ESI basis as the model's anchor; a cut-detection bug (string vs integer) was caught and fixed before shipping. |

ROADMAP

# v13 is the fall arc: height, mortality, and the ask

| | | |
|---|---|---|
| 1 | **v13.0 Height and domain** | Gate off predictions outside the ESI range; blend sweep parked |
| 2 | **v13.1 Shipping decision** | ship18 scored on all ten sources and, as of 15 Aug, on the pooled composite: 85.4 (NS) / 106.5 (NS-long), tied with base, behind v12.11 and OSM; its gains are concentrated on the developing Maine FIA target (-16.6 to -12.5). Disclosed tradeoffs: Reineke softwood slope -1.80 exactly on the acceptance boundary; small-tree fix roughly 1.2 pp BA cost at Maine FIA. Still blocked on biological realism: the full 3.9x ingrowth stack and SDImax removed. Next: reparameterize the basal-area increment site response, then rescore. |
| 3 | **The ask of the group** | Provincial managed remeasurements, PSP treatment histories, and site surfaces |

VERDICT AND KEY TAKEAWAYS

# v12.11 is ready, and its open gap is named

- 1 With every new correction switched off, FVS-ACD reproduces the shipped model exactly. Every number traces to a script.
- 2 Managed CTRN, NB MagPlot, and mature Baxter: v12.11 halves bias (Baxter +11.28 vs OSM +16.85, corrected 15 Aug; CTRN -2.24 vs OSM -4.81; NB -5.96 vs OSM -8.07). OSM keeps the lower pooled %RMSE on common cells and is unscored on two of eight sources.
- 3 Developing Maine FIA: OSM leads (-6.3% vs -16.1%); v13.1 (ship18) narrows FVS-ACD to -12.5% and does not close it
- 4 Nova Scotia bias weakens, never flips positive (managed long horizon: v12.11 -1.66% vs OSM -9.14%)
- 5 v13.1 (ship18): composite 85.4 / 106.5 ties base and trails v12.11 and OSM; its gains sit on the Maine FIA target; one self-thinning flag (-1.80) disclosed

# Questions / Comments

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*AcadianGY is open R, v12.6.1 to v12.11 now git tagged; plot coordinates stay on the analysis cluster, aggregates only.*

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